AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	81.0 / Male
Department	Urology
Diagnosis	Severe kidney infection
UTI Classification	Complicated UTI
Sample Type	Urine

Clinical Presentation

Chief Complaints: Fever;Flank pain;Vomiting
Comorbidities: Diabetes;Hypertension
Surgical History: Prostate surgery
Social History: Ex smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	23.0	%	20 - 40
Wbc	21800.0	cells/µL	4000 - 11000
Polymorphs	85.0	%	40 - 75
Crp	100.0	mg/L	< 10
Rft Serum Creatinine	3.0	mg/dL	0.6 - 1.3
Serum Uric Acid	6.9	mg/dL	3.5 - 7.2
Blood Urea	64.0	mg/dL	7 - 20
Cue Pus Cells	34.0	/HPF	0 - 5
Epithelial Cells	6.0	/HPF	0 - 5
Proteins	3+		Negative
Rbc	7.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	**Proteus vulgaris**
Bacteria Type	Gram-negative bacillus (Indole-positive; more resistant than P. mirabilis)

Antibiotic Susceptibility Profile

Predicted Sensitive	**Amikacin, Imipenem, Piperacillin-Tazobactam**
Predicted Resistant	**Cefixime**

Recommended Therapy

Imipenem-cilastatin

Dosage: 500 mg IV every 8 hours (adjusted for estimated CrCl ~30 mL/min)

Precautions: Renal dose adjustment required; monitor serum creatinine and drug levels; avoid in patients with known carbapenem allergy; watch for seizures in elderly with high CNS penetration.

Rationale: Imipenem is a broad-spectrum carbapenem with excellent activity against Proteus vulgaris, including indole-positive, more resistant strains. It is one of the few agents in the sensitive list that can be used safely with renal impairment when dose-adjusted, making it a primary choice for this severe, complicated kidney infection.

Piperacillin-tazobactam

Dosage: 3.375 g IV every 8 hours (adjusted for reduced renal function)

Precautions: Renal dose adjustment required; monitor renal function and watch for possible hypersensitivity reactions; caution in patients with a history of beta-lactam allergy.

Rationale: Piperacillin-tazobactam provides reliable coverage against Gram-negative bacilli such as Proteus vulgaris and penetrates renal tissue well. It is less nephrotoxic than aminoglycosides and can be used with dose reduction in this patient with elevated creatinine.

Antibiotic Drug History

Clinical Summary

Infection Overview

• **Type:** Complicated urinary tract infection - emphysematous pyelonephritis of the kidney.

• **Likely pathogen:** *Proteus vulgaris* (Gram-negative bacillus, indole-positive, more resistant than *P. mirabilis*).

Resistance / Sensitivity Profile

• **Predicted resistant:** Cefixime (previously used).

• **Predicted susceptible:** Amikacin, Imipenem, Piperacillin-tazobactam.

Recommended Antimicrobial Therapy

1. **Imipenem-cilastatin** 500 mg IV q8 h

• Adjust dose for estimated CrCl ≈30 mL/min (renal dose reduction).

• **Rationale:** Carbapenem offers the broadest activity against the resistant *P. vulgaris* strain and retains efficacy despite renal impairment when dose-adjusted. It is a first-line choice for severe, tissue-penetrating infections such as emphysematous pyelonephritis.

2. **Piperacillin-tazobactam** 3.375 g IV q8 h

• Renal dose adjustment required.

• **Rationale:** Provides reliable coverage of Gram-negative bacilli, good renal tissue penetration, and is less nephrotoxic than aminoglycosides-important given the patient's creatinine of 3 mg/dL.

Key Precautions & Monitoring

• **Renal function:** Daily serum creatinine and estimated GFR; adjust doses accordingly.

• **Allergy:** Verify no history of carbapenem or β-lactam allergy; monitor for hypersensitivity reactions.

• **Neurotoxicity:** Elderly patients are at higher risk for seizures with imipenem; observe mental status.

• **Drug interactions:** Review concomitant meds for potential interactions with carbapenems or β-lactamase inhibitors.

Actionable Plan

Start the above regimen promptly, obtain repeat cultures within 48 h, and reassess therapy based on susceptibility results and renal trends. Consider de-escalation if a narrower-spectrum agent becomes appropriate.